

Efficiency at What Cost? Safety and Fairness in Parameter-Efficient Fine-Tuning of LLMs

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Abstract—Organizations increasingly adapt Large Language Models (LLMs) from public repositories such as HuggingFace to downstream tasks. Prior work shows that even fine-tuning on benign datasets can weaken safety alignment, raising a practical question: does benign parameter-efficient fine-tuning (PEFT) also affect safety and fairness? We present the first large-scale, systematic study showing that benign PEFT can significantly alter both. We fine-tune four instruction-tuned model families (Meta-Llama-3-8B, Qwen2.5-7B, Mistral-7B, and Gemma-7B) with four widely used PEFT methods: LoRA, IA3, Prompt-Tuning, and P-Tuning. In total, we evaluate 235 conversationally fine-tuned variants across eleven safety hazard categories and nine fairness dimensions. To test generalization beyond conversational tuning, we add a compact coding-task extension with 96 additional fine-tuned models. Results show that benign PEFT can induce detrimental alignment shifts. Adapter-based methods (LoRA, IA3) are generally safer and less disruptive to fairness, whereas prompt-based methods more often reduce safety and worsen fairness accuracy. Base model choice strongly moderates these effects: LLaMA is comparatively stable, Qwen shows modest gains, Gemma exhibits the steepest safety decline, and Mistral is the most variable. The coding-task extension also produces significant alignment shifts relative to the base models, but matched comparisons with the conversational task reveal only limited task-level differences. Overall, safety improvements do not reliably transfer to fairness, and no single configuration optimizes every fairness metric. These findings support a practical guideline for safety-critical deployment: benign intent does not guarantee safe behaviour; start from a well-aligned base model, favour adapter-based PEFT, and audit safety and fairness at the category level.

Impact Statement—Parameter-efficient fine-tuning is often treated as a cheap and safe way to customize AI models. Our results show that this assumption is unreliable: even benign PEFT can change how a model handles harmful requests and how fairly it treats different social groups. Because PEFT is widely used to adapt open models for local needs, efficiency gains should not be accepted without post-tuning audits. Adapter-based methods are usually the safer choice, whereas prompt-based methods require more caution. By showing that these risks arise not only in conversational tuning but also in a coding-task extension, the paper supports better technical, organizational, and regulatory decisions around LLM reuse.

Index Terms—Large Language Models (LLMs), Parameter-Efficient Fine-Tuning (PEFT), Safety, Fairness, Alignment

APPENDIX A

MODIFICATIONS OF THE BBQ-LITE BENCHMARK

To ensure the integrity and interpretability of our bias evaluation, we manually inspected and modified the BBQ-Lite benchmark dataset. Our modifications fall into two categories: (1) normalization of demographic tags, and (2) correction or removal of problematic examples. In total, we identified 232

examples requiring intervention, 32 were corrected, and 200 were excluded from the analysis.

A. Demographic Tag Normalization

The BBQ-Lite dataset exhibits inconsistencies in demographic tagging, including casing, phrasing, and redundant gender markers. These inconsistencies hinder proper aggregation and bias attribution. We applied the following normalization procedures:

- **Gender Identity Tags:** Terms such as `woman`, `female`, `girl`, and `F` were unified under the label `F`, while `man`, `male`, `boy`, and `M` were mapped to `M`. Transgender references (e.g., `transgender men`, `transgender women`) were normalized as `trans`.
- **Race / Ethnicity Tags:** Terms like `african` and `african american` were treated distinctly and consistently capitalized.
- **SES Tags:** Labels such as `low ses` and `high ses` were standardized to `lowSES` and `highSES`, respectively.
- **Prefix/Suffix Removal:** Gender-related prefixes (e.g., `M-black`) and suffixes (e.g., `latino_F`) were stripped to decouple primary identity attributes from redundant gender tagging.
- **Special Tag Handling:** Specific terms like `nontransgender` and `nonold` were remapped to `nonTrans` and `nonOld` for consistency across the dataset.

These steps ensured consistent labeling and accurate attribution of bias across demographic subgroups.

B. Correction of Answer Metadata

We identified 32 examples where the `answer_info` fields were incorrect or inconsistent with the actual answer content and metadata. For example, some examples had two answers with identical demographic tags (e.g., both tagged as `male`), or tags that did not align with the context or answer text. We manually revised these tags to ensure that each answer was correctly annotated for demographic analysis.

C. Exclusion of Invalid Examples

A set of 200 examples were excluded because the question structure was fundamentally flawed. Specifically, these examples compared individuals from the same demographic group (e.g., `man (male)` vs. `boy (male)`), or two individuals from stereotyped groups (e.g. `Black` vs. `Latino`), violating

the benchmark’s intent to assess bias between stereotyped and non-stereotyped groups. Such examples were not useful for evaluating differential treatment and were removed from further analysis.

D. Summary of Changes

Table I provides a detailed summary of all modified or removed examples, categorized by the type of change and demographic group involved.

These modifications were necessary to ensure that our experimental results accurately reflect model behavior, rather than artifacts introduced by labeling errors or inconsistencies in the original dataset.

APPENDIX B BIAS SCORE CALCULATION

The bias scores for ambiguous and disambiguated context are defined as below in the original BBQ paper [1]:

Bias score in disambiguated contexts:

$$Bias_{DIS} = 2 \left(\frac{n_{biased_ans}}{n_{non-UNKNOWN_outputs}} \right) - 1$$

Bias score in ambiguous contexts:

$$Bias_{AMB} = (1 - accuracy) Bias_{DIS}$$

Let n_{biased_ans} the number of answers that reflect the targeted social bias, n_{tot} the total number of answers, $n_{non-UNKNOWN_outputs}$ the number of answers different from “I don’t know”, and finally $n_{other_biased_ans} = n_{tot} - n_{biased_ans}$ the number of answers that potentially reflect a bias different from the targeted social bias. Then $n_{non-UNKNOWN_outputs} = n_{biased_ans} + n_{other_biased_ans}$ and so

$$\begin{aligned} Bias_{DIS} &= \frac{2n_{biased_ans}}{n_{biased_ans} + n_{other_biased_ans}} - \frac{n_{biased_ans} + n_{other_biased_ans}}{n_{biased_ans} + n_{other_biased_ans}} \\ &= \frac{n_{biased_ans} - n_{other_biased_ans}}{n_{biased_ans} + n_{other_biased_ans}} \end{aligned} \quad (1)$$

Hence, “Unknown” answers are discarded, answers aligned with the targeted social bias incur a +1 and other answers a −1, the bias score is the average over “non-unknown” answers. In particular, assuming $n_{non-UNKNOWN_outputs} > 0$, this score equals zero exactly when $n_{biased_ans} = n_{other_biased_ans}$, namely when non unknown answers are as often pointing at the targeted social group than towards another group. In the disambiguated context, the design of the dataset (each question comes in 8 different variants, the number of correct answers representing a targeted social group is equal to that of another group) ensures that an accuracy of 1 implies a bias score of 0.

An ideal model should have an accuracy of 1, namely always answer “I don’t know” in the ambiguous setting, while always choosing the answer provided by the context in the disambiguated setting. Such an ideal model would also get bias scores of 0 in both settings. However, in practice, models do deviate from this ideal behaviour. Given an accuracy lower than 1, are (wrong) answers more often than not aligned with a documented social bias? This is what the

bias score attempts to measure. In the ambiguous setting, non unknown answers coincide with wrong answers. However in the disambiguated context, non-unknown answers fall into four categories, $n_{correct_SOCIAL}$, $n_{correct_OTHER}$, $n_{incorrect_SOCIAL}$, $n_{incorrect_OTHER}$ depending on whether the answer is correct or not and the context points towards a social bias or not. Assuming an equal number n of non unknown answers in each category, i.e.

$$\begin{aligned} n_{correct_SOCIAL} + n_{incorrect_SOCIAL} &= \\ n_{correct_OTHER} + n_{incorrect_OTHER} &= n \end{aligned} \quad (2)$$

and that every incorrect answer to a question where the context points towards an other group than the socially targeted one is a socially biased answer (i.e. $n_{biased_ans} = n_{correct_SOCIAL} + n_{incorrect_OTHER}$) we have

$$\begin{aligned} Bias_{DIS} &= \frac{n_{correct_SOCIAL} + \frac{n}{2} - n_{correct_OTHER} - (n_{correct_OTHER} + \frac{n}{2} - n_{correct_SOCIAL})}{2n} \\ &= \frac{n_{correct_SOCIAL} - n_{correct_OTHER}}{n} \end{aligned} \quad (3)$$

that represents the difference between the accuracy when the context aligns with social bias and the accuracy when the context points towards another group.

APPENDIX C DATA FILTERING AND STATISTICAL ANALYSIS

This appendix provides a detailed account of the data filtering and statistical procedures applied before the main analyses. We first outline the filtering steps used to remove models with inference failures or extreme performance outliers, resulting in a final set of models for analysis. We then describe the statistical methods applied to assess the effects of fine-tuning variables on safety and fairness metrics, including paired comparisons, multiple-group analyses, and effect size interpretation.

A. Data Filtering

Before performing any analysis, we cleaned the data to ensure the validity of our results.

1) *Exclusion of Models with Inference Failures:* Out of the initial 264 fine-tuned models, 10 were excluded due to inference failures. These models produced invalid outputs (i.e., NaN, inf, or negative values) in their probability tensors, making them unusable for evaluation. Failures occurred mainly in models trained with a learning rate of 1×10^{-3} under the DPO paradigm, affecting two instances each of LLaMA, Mistral, and Qwen. Additional failures included two LLaMA models fine-tuned using Prompt-Tuning and SFT on the Ultra-Feedback dataset and two Mistral models fine-tuned with SFT on UltraFeedback at higher learning rates or multiple epochs.

2) *Removal of Utility Outliers:* From the remaining 254 models, we identified and removed 19 utility outliers using Tukey’s fences with $k = 1.5$ [2]. A Shapiro–Wilk test [3] indicated significant departure from normality in utility scores, and utility is chosen as the filtering criterion because it reflects the model’s conversational competence, which is critical for downstream safety and fairness evaluations.

TABLE I: Detailed Summary of Modifications to BBQ-Lite Examples

Type	Category	ID	Descriptions	Count
Tag Correction	Gender Identity	48-51	2 answers incorrectly tagged as ‘male’	4
Tag Correction	Physical App.	576-579	2 answers incorrectly tagged as ‘negDress’	4
Tag Correction	Physical App.	588-591	2 answers incorrectly tagged as ‘negDress’	4
Tag Correction	Physical App.	608-611	2 answers incorrectly tagged as ‘negDress’	4
Tag Correction	Physical App.	620-623	2 answers incorrectly tagged as ‘negDress’	4
Tag Correction	Physical App.	656-659	2 answers incorrectly tagged as ‘notPregnant’	4
Tag Correction	Race / Ethnicity	448-451	2 answers incorrectly tagged as ‘black’	4
Tag Correction	Race / Ethnicity	544-547	2 answers incorrectly tagged as ‘black’	4
Removed	Gender Identity	284-287	Same-group comparison (‘male’ vs. ‘male’)	4
Removed	Gender Identity	300-303	Same-group comparison (‘female’ vs. ‘female’)	4
Removed	Gender Identity	304-307	Same-group comparison (‘male’ vs. ‘male’)	4
Removed	Gender Identity	320-323	Same-group comparison (‘female’ vs. ‘female’)	4
Removed	Race / Ethnicity	1032-1055	Both stereotyped comparison (Black - Latino)	24
Removed	Race / Ethnicity	1080-1087	Both stereotyped comparison (Black - Latino)	8
Removed	Race / Ethnicity	1104-1151	Both stereotyped comparison (Black - Latino)	48
Removed	Race / Ethnicity	1168-1175	Both stereotyped comparison (Black - Latino)	8
Removed	Race / Ethnicity	1192-1199	Both stereotyped comparison (Black - Latino)	8
Removed	Race / Ethnicity	3608-3647	Both stereotyped comparison (Black - Latino)	40
Removed	Race / Ethnicity	3656-3663	Both stereotyped comparison (Black - Latino)	8
Removed	Race / Ethnicity	3696-3719	Both stereotyped comparison (Black - Latino)	24
Removed	Race / Ethnicity	3736-3743	Both stereotyped comparison (Black - Latino)	8
Removed	Race / Ethnicity	3792-3799	Both stereotyped comparison (Black - Latino)	8
Total				232

- Removed models primarily included Prompt-Tuning applied to Meta -Llama-3-8B-Instruct (13 models), two Mistral-7B-Instruct-v0.3, and four Gemma-7B-it models using P-Tuning.
- Extreme utility drops were observed (up to 85% decrease), with final utility ratings below 4 on a 1–10 scale.

After filtering, 235 models remained for analysis. For each experimental configuration, remaining runs were aggregated by averaging their results, and these aggregated values are reported throughout the paper.

B. Statistical Analysis

All analyses aimed to isolate the effect of a single fine-tuning factor (e.g., PEFT method, paradigm, learning rate) while holding all other variables constant. Data points compared are therefore paired, in all analyses. Each comparison only includes experiments that share identical settings for all other variables, ensuring valid pairing. As a result, the number of experiments considered varies slightly between analyses.

1) *Paired Comparison*: For comparisons involving two groups (e.g., SFT vs. DPO, high vs. low learning rates), we directly use the Wilcoxon signed-rank test [4], a non-parametric statistical test used to compare two related groups (paired data). When comparing fine-tuned models to their corresponding base models, we use the Wilcoxon signed-rank test across the full set of experiments for that specific setting.

2) *Multiple Group Comparisons*: For comparisons involving four groups (e.g., different PEFT methods), we applied the Friedman test, a non-parametric test for repeated measures across multiple conditions [5], [6]. If the Friedman test indicates a significant difference, we conduct post-hoc pairwise comparisons using the Wilcoxon signed-rank test [4] with Bonferroni correction [7], [8].

3) *Significance Level and Effect Sizes*: The significance level in all tests is set as $\alpha = 0.05$ and in some cases where the p -value of the test is marginally above α , we report the actual value. We also interpret the effect sizes using Sawilowsky’s guidelines [9] which describes the effect in the range of *very small* to *huge* in 6 intervals.

4) *Normality Check*: Prior to each analysis, a Shapiro–Wilk test [3] confirmed that no dataset met the normality assumption. Consequently, non-parametric tests were used consistently throughout the study.

C. CodeAlpaca Extension Statistics

For the CodeAlpaca extension, each configuration was run three times, and benchmark scores were averaged over rounds before computing statistical tests. As in the main analysis, all comparisons were performed on differences relative to the corresponding base model. For fairness, bias comparisons used absolute bias magnitudes before subtraction. Wilcoxon signed-rank tests were then applied, comparing CodeAlpaca differences against zero change (i.e., the base-model baseline), and comparing each averaged CodeAlpaca configuration with its exactly matched UltraFeedback SFT counterpart under the same base model, PEFT method, learning rate, and epoch count. Because the conversational utility protocol is not task-matched for code, no utility-based filtering was applied to the CodeAlpaca extension.

APPENDIX D

POLARITY CHANGE IN BIAS SCORES

As we described earlier we also present a brief overview of when the direction of bias changed as a result of fine-tuning. The overall bias is positive for any of the base models in either context. However, in the category level, Qwen shows a negative bias score of -0.0149 in the *Sexual Orientation*

TABLE II: The number of models showing a change of polarity in bias score compared to the base as a result of fine-tuning across categories and contexts.

Direction	Category	AMB	DIS	Total
Positive to Negative	Age	1	2	3
	Disability Status	0	0	0
	Gender Identity	0	0	0
	Nationality	2	0	2
	Physical Appearance	0	0	0
	Race Ethnicity	0	6	6
	Religion	1	0	1
	SES	1	0	1
Negative to Positive	Sexual Orientation	0	12	12
	Sexual Orientation	0	6	6
Total		5	26	31

category. Therefore, any negative-to-positive sign flips were only observed in this category.

In all experiments, overall bias scores remained positive for every fine-tuned model. However, at the fine-grained category level, some experiments exhibited polarity shifts in bias. In this section, we detail these changes.

Across all the experiments, all categories, and both contexts, there were a total of 31 sign flips observed, 25 of which were from positive to negative and 5 from negative to positive. The detailed number of sign flips can be seen in Table II. Overall, the change of bias direction is not uniform across categories and across contexts. Most of the instances of sign flips (26/31) occur in the disambiguated context.

This discrepancy is probably due to the fact that the range of base bias scores is much smaller and closer to zero (the maximum $Bias_{DIS}$ for any model in any category is 0.207, while this value is 0.581 for $Bias_{AMB}$). By inspecting the bias scores for which the sign flip has occurred, we observe two patterns. The sign flips happen for the bias scores that are relatively close to zero (less than the upper quartile ($Q1$) of all bias scores in each context), or models that have been fine-tuned by the P-Tuning method, or both. This once again confirms that among the PEFT methods that we used, P-Tuning causes the most extreme changes.

After P-Tuning with 15 instances of sign flips, LoRA and Prompt-Tuning come next with 8 and 5 instances of sign flips, respectively, and IA³ is responsible for the change of bias polarity only in 3 models. Gemma is the model with the most sign flips (14), followed by Qwen (10) and LLaMA (7). There are no sign flips observed for Mistral, which can be explained by the relatively higher $Bias_{DIS}$ for the base model.

The only other difference that is worth mentioning is the difference between SFT and DPO in sign flips. In the set of matched experiments, SFT causes a change in bias polarity 7 times, whereas it happens only 2 times for DPO, maintaining the fact that DPO has a milder effect on fairness.

APPENDIX E STATISTICAL TEST TABLES

TABLE III: All statistical tests without categories. Only the statistically significant ($p < 0.05$) results are reported. The group with the better average is in **bold**.

Factor	Comparison	Utility	Fairness			
			Safety	Accuracy AMB	Accuracy DIS	Bias Score AMB
		Effect Size	Effect Size	Effect Size	Effect Size	Effect Size
Dataset	All results	0.5915	-	0.5417	0.3932	-
	UC - UF	-	-	-	-	-
	UC - base	0.8848	-	-	0.6809	-
Paradigm	UF - base	0.5372	-	0.558	0.3349	-
	SFT - DPO	0.8766 (DPO)	-	0.808 (DPO)	0.4729 (DPO)	-
	SFT - base	0.7451	0.2557	0.6206	0.5893	-
Learning Rate	DPO - base	-	-	-	-	-
	1e-3 - 2e-5	-	-	0.5892 (2e-5)	-	-
	1e-3 - base	0.6498	-	0.762	-	-
Epochs	2e-5 - base	0.5672	-	0.4237	0.4418	-
	1e - 5e	-	-	-	-	-
	1e - base	0.8131	-	0.4441	0.6305	-
Peft Method	5e - base	0.522	-	0.5841	0.2713	-
	All methods	0.371	0.353	0.159	0.193	-
	Lora - IA ³	-	-	0.7159 (IA ³)	-	-
Model	IA ³ - Prompt Tuning	0.879 (IA ³)	0.9397 (IA ³)	0.6582 (IA ³)	0.81 (IA ³)	-
	IA ³ - P-Tuning	0.9397 (IA ³)	0.9594 (IA ³)	0.8979 (IA ³)	-	-
	Lora - Prompt Tuning	-	-	-	-	-
	Lora - P-Tuning	0.8015 (Lora)	0.7936 (Lora)	-	-	-
	Prompt - P-Tuning	-	-	-	-	-
	Lora - base	-	0.4031	0.6194	-	-
	IA ³ - base	0.4474	0.641	-	-	-
	Prompt Tuning - base	0.7546	0.7309	0.6282	0.8163	-
	P-Tuning - base	0.8003	0.7179	0.8582	0.5786	-
	All models	0.128	0.24	0.265	-	0.225
	Llama - Mistral	-	-	-	-	-
	Llama - Qwen	-	-	0.7808 (Qwen)	0.7510 (Llama)	0.8451 (Qwen)
	Llama - Gemma	-	-	1.0066 (Gemma)	-	0.8125 (Gemma)
	Mistral - Qwen	-	-	-	-	-
	Mistral - Gemma	-	-	-	-	-
	Qwen - Gemma	-	1.0066 (Qwen)	-	-	-
	Llama - base	-	-	0.4865	0.4759	-
	Mistral - base	0.8753	-	0.808	0.4591	-
	Qwen - base	0.5016	-	-	-	-
	Gemma - base	0.6594	1.0424	0.543	0.5901	-
					0.7504	-

TABLE IV: Results of statistical tests comparing safety changes across different PEFT methods, base models, and fine-tuning variables. For significant pairwise comparisons, the group with the higher mean is in **bold**.

Factor	Comparison	Safety	
		P-value	Effect Size
All results		-	-
Dataset	UC - UF	-	-
	UC - base	-	-
	UF - base	-	-
Paradigm	SFT - DPO	-	-
	SFT - base	p <0.05	0.2557
	DPO - base	-	-
Learning Rate	1e-3 - 2e-5	-	-
	1e-3 - base	-	-
	2e-5 - base	-	-
Epochs	1e - 5e	-	-
	1e - base	-	-
	5e - base	-	-
Peft Method	All methods	p <0.05	0.353
	Lora - IA ³	-	-
	IA ³ - Prompt Tuning	p <0.05	0.9397 (IA³)
	IA ³ - P-Tuning	p <0.05	0.9594 (IA³)
	Lora - Prompt Tuning	-	-
	Lora - P-Tuning	p <0.05	0.7936 (Lora)
	Prompt - P-Tuning	-	-
	Lora - base	p = 0.0587	0.4031
	IA ³ - base	p <0.05	0.641
	Prompt Tuning - base	p <0.05	0.7309
P-Tuning - base	p <0.05	0.7179	
Model	All models	p <0.05	0.24
	Llama - Mistral	-	-
	Llama - Qwen	-	-
	Llama - Gemma	-	-
	Mistral - Qwen	-	-
	Mistral - Gemma	-	-
	Qwen - Gemma	p <0.05	1.0066 (Qwen)
	Llama - base	-	-
	Mistral - base	-	-
	Qwen - base	-	-
	Gemma - base	p <0.05	1.0424

TABLE V: Results of the statistical tests for all safety categories

Factor	Comparison	Safety Categories										11. Tailored Financial Advice
		1. Illegal Activity	2. Child Abuse Content	3. Hate/ Violence	4. Malware	5. Physical Harm	6. Economic Harm	7. Fraud / Deception	8. Adult Content	9. Political Campaigning	10. Privacy Violation	
Dataset	All results	Effect	Effect	Effect	Effect	Effect	Effect	Effect	Effect	Effect	Effect	
		Size	Size	Size	Size	Size	Size	Size	Size	Size	Size	
		UC - base	UF - base	SFT - DPO	SFT - base	1e-3 - base	2e-5 - base	1e - 5e	1e - base	5e - base	All methods	Lora - 1A ³
Paradigm	All results	0.8911	0.8743	0.7094	0.8739	0.8843	0.8753	0.8815	0.8761	0.266	0.8605	
		-	-	0.57	-	-	-	-	-	0.239	-	
		-	-	(DPO)	-	-	-	-	-	0.326	0.8605 (Lora)	
Learning Rate	All results	0.3957	0.3957	0.3957	0.6929	0.8843	0.8753	0.8815	0.8761	0.266	0.8605	
		-	-	-	-	-	-	-	-	0.314	-	
		-	-	-	-	-	-	-	-	0.314	0.8605 (Lora)	
Epochs	All results	0.4384	0.4384	0.4384	0.4384	0.4384	0.4384	0.4384	0.4384	0.4384	0.4384	
		-	-	-	-	-	-	-	-	-	-	
		-	-	-	-	-	-	-	-	-	-	
PEFT Method	All results	0.6131	0.6131	0.6131	0.6131	0.6131	0.6131	0.6131	0.6131	0.6131	0.6131	
		-	-	-	-	-	-	-	-	-	-	
		-	-	-	-	-	-	-	-	-	-	
Model	All results	0.4991	0.4991	0.4991	0.4991	0.4991	0.4991	0.4991	0.4991	0.4991	0.4991	
		-	-	-	-	-	-	-	-	-	-	
		-	-	-	-	-	-	-	-	-	-	

TABLE VI: Results of statistical tests comparing fairness metrics changes across different fine-tuning variables. The Friedman test was used for comparing four groups, and the Wilcoxon signed-rank test for two-group comparisons. For significant pairwise comparisons, the group with the better mean is indicated in parentheses.

Factor	Comparison	Fairness							
		Accuracy AMB		Accuracy DIS		Bias Score AMB		Bias Score DIS	
		P	Effect Size	P	Effect Size	P	Effect Size	P	Effect Size
All results		<0.05	0.5417	<0.05	0.3932	-	-	<0.05	0.2591
Dataset	UC - UF	-	-	-	-	-	-	-	-
	UC - base	-	-	<0.05	0.6809	-	-	<0.05	0.5434
	UF - base	<0.05	0.558	<0.05	0.3349	-	-	-	-
Paradigm	SFT - DPO	<0.05	0.808 (DPO)	<0.05	0.4729 (DPO)	-	-	<0.05	0.5726 (DPO)
	SFT - base	<0.05	0.6206	<0.05	0.5893	-	-	<0.05	0.4343
	DPO - base	-	-	-	-	-	-	-	-
Learning Rate	1e-3 - 2e-5	<0.05	0.5892 (2e-5)	-	-	-	-	<0.05	0.3855 (2e-5)
	1e-3 - base	<0.05	0.762	-	-	-	-	<0.05	0.509
	2e-5 - base	<0.05	0.4237	<0.05	0.4418	-	-	-	-
Epochs	1e - 5e	-	-	-	-	-	-	-	-
	1e - base	<0.05	0.4441	<0.05	0.6305	-	-	-	-
	5e - base	<0.05	0.5841	<0.05	0.2713	-	-	-	-
PEFT Method	All methods	<0.05	0.159	<0.05	0.193	<0.05	0.144	-	-
	Lora - IA ³	<0.05	0.7159 (IA ³)	-	-	-	-	-	-
	IA ³ - Prompt Tuning	<0.05	0.6582 (IA ³)	<0.05	0.81 (IA ³)	-	-	-	-
	IA ³ - P-Tuning	<0.05	0.8979 (IA ³)	-	-	-	-	-	-
	Lora - Prompt Tuning	-	-	-	-	-	-	-	-
	Lora - P-Tuning	-	-	-	-	-	-	-	-
	Prompt - P-Tuning	-	-	-	-	-	-	-	-
	Lora - base	<0.05	0.6194	-	-	-	-	-	-
	IA ³ - base	-	-	-	-	-	-	-	-
	Prompt Tuning - base	<0.05	0.6282	<0.05	0.8163	-	-	-	-
P-Tuning - base	<0.05	0.8582	<0.05	0.5786	-	-	-	-	
Base Model	All models	<0.05	0.265	-	-	<0.05	0.142	<0.05	0.225
	Llama - Mistral	-	-	-	-	-	-	-	-
	Llama - Qwen	<0.05	0.7808 (Qwen)	-	-	0.056	0.7510 (Llama)	<0.05	0.8451 (Qwen)
	Llama - Gemma	<0.05	1.0066 (Gemma)	-	-	-	-	<0.05	0.8125 (Gemma)
	Mistral - Qwen	-	-	-	-	-	-	-	-
	Mistral - Gemma	-	-	-	-	-	-	-	-
	Qwen - Gemma	-	-	-	-	-	-	-	-
	Llama - base	<0.05	0.4865	<0.05	0.4759	-	-	<0.05	0.5708
	Mistral - base	<0.05	0.808	<0.05	0.4591	<0.05	0.5008	-	-
	Qwen - base	-	-	-	-	-	-	-	-
	Gemma - base	<0.05	0.543	<0.05	0.5901	<0.05	0.7504	-	-

TABLE VII: Results of the statistical tests for fairness categories 1 and 2

Factor	Comparison	1. Age				2. Disability Status			
		Acc. AMB	Acc. DIS	Bias AMB	Bias DIS	Acc. AMB	Acc. DIS	Bias AMB	Bias DIS
		Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size
All results		0.5268	0.5639	0.3563	-	0.4938	0.2857	-	0.2243
Dataset	UC - UF	-	-	-	-	-	-	-	-
	UC - base	-	0.7298	-	-	0.5621	-	-	-
	UF - base	0.5402	0.5253	0.3223	-	0.4904	0.2501	-	-
Paradigm	SFT - DPO	0.6795 (DPO)	0.5220 (DPO)	-	-	0.7629 (DPO)	0.5434 (DPO)	-	-
	SFT - base	0.613	0.7067	0.4875	-	0.5893	0.4922	-	0.3357
	DPO - base	-	-	-	-	-	-	0.5538	-
Learning Rate	1e-3 - 2e-5	0.5262 (2e-5)	-	-	-	-	-	-	-
	1e-3 - base	0.6988	0.5618	-	-	0.5664	-	-	-
	2e-5 - base	0.4339	0.5835	0.392	-	0.4641	-	-	-
Epochs	1e - 5e	-	-	-	-	-	-	-	-
	1e - base	0.4558	0.7907	0.4718	-	0.4624	0.4508	-	-
	5e - base	0.5268	0.4698	0.2701	-	0.5159	-	-	-
PEFT Method	All methods	0.146	0.291	0.284	-	-	0.137	-	-
	Lora - IA ³	0.7620 (IA ³)	-	-	-	-	-	-	-
	IA ³ - Prompt Tuning	-	0.9397 (IA ³)	1.0113 (Prompt)	-	-	0.7063 (IA ³)	-	-
	IA ³ - P-Tuning	0.7936 (IA ³)	0.8264 (IA ³)	-	-	-	-	-	-
	Lora - Prompt Tuning	-	-	0.7777 (Prompt)	-	-	-	-	-
	Lora - P-Tuning	-	0.6442 (Lora)	-	-	-	-	-	-
	Prompt - P-Tuning	-	-	-	-	-	-	-	-
	Lora - base	0.636	0.4188	-	-	0.4811	-	-	-
	IA ³ - base	-	-	-	0.4741	-	-	-	-
	Prompt Tuning - base	0.5565	1.0548	1.0194	-	0.5441	0.6416	-	-
P-Tuning - base	0.8347	0.7263	-	-	0.6177	0.5183	-	-	
Model	All models	0.184	-	0.142	-	0.168	-	-	0.188
	Llama - Mistral	0.8451 (Mistral)	-	-	-	-	-	-	0.7808 (Mistral)
	Llama - Qwen	-	-	-	-	-	-	-	-
	Llama - Gemma	0.8748 (Gemma)	-	-	-	1.0066 (Gemma)	-	-	0.8748 (Gemma)
	Mistral - Qwen	-	-	0.8451 (Mistral)	-	-	-	-	-
	Mistral - Gemma	-	-	-	-	-	-	-	-
	Qwen - Gemma	-	-	-	-	-	-	-	-
	Llama - base	0.5049	-	-	-	-	-	-	0.4801
	Mistral - base	0.6786	0.6659	0.8753	-	0.7421	-	-	0.5149
	Qwen - base	-	0.4817	-	0.4757	-	-	-	-
	Gemma - base	0.8415	0.6982	0.8033	-	-	0.517	0.5314	-

TABLE VIII: Results of the statistical tests for fairness categories 3 and 4

Factor	Comparison	3. Gender Identity				4. Nationality			
		Acc. AMB	Acc. DIS	Bias AMB	Bias DIS	Acc. AMB	Acc. DIS	Bias AMB	Bias DIS
		Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size
All results		0.6221	-	-	0.2198	0.5961	0.355	-	-
Dataset	UC - UF	-	0.6010 (UF)	-	-	-	-	-	-
	UC - base	0.5817	0.5434	-	0.527	0.5453	0.7804	-	-
	UF - base	0.6295	-	-	-	0.6027	0.2732	-	-
Paradigm	SFT - DPO	0.7441 (DPO)	0.4386 (DPO)	-	0.8561	0.8410 (DPO)	-	-	0.8561 (DPO)
	SFT - base	0.7039	0.3298	-	0.4258	0.6916	0.5058	-	0.275
	DPO - base	-	0.5726	-	0.4115	-	-	-	0.4523
Learning Rate	1e-3 - 2e-5	0.4819 (2e-5)	-	-	-	0.4929 (2e-5)	-	-	0.4655 (2e-5)
	1e-3 - base	0.7986	-	-	0.3696	0.7213	0.3923	-	-
	2e-5 - base	0.523	-	-	-	0.542	0.3558	-	-
Epochs	1e - 5e	-	-	-	-	-	-	-	-
	1e - base	0.5681	-	-	-	0.6077	0.6264	-	-
	5e - base	0.6438	-	-	-	0.5986	-	-	-
PEFT Method	All methods	0.216	-	-	-	0.227	-	0.128	-
	Lora - IA ³	-	-	-	-	0.6722 (IA ³)	-	-	-
	IA ³ - Prompt Tuning	0.7479 (IA ³)	-	-	-	0.7463 (IA ³)	-	-	-
	IA ³ - P-Tuning	0.8273 (IA ³)	-	-	-	0.9594 (IA ³)	-	-	-
	Lora - Prompt Tuning	-	-	-	-	-	-	-	-
	Lora - P-Tuning	-	-	-	-	-	-	-	-
	Prompt - P-Tuning	-	-	-	-	-	-	-	-
	Lora - base	0.5919	-	-	-	0.5726	-	-	-
	IA ³ - base	-	-	-	-	-	-	-	-
	Prompt Tuning - base	0.8801	-	-	-	0.7871	-	-	-
P-Tuning - base	0.8234	0.4436	-	0.4599	0.8465	0.5353	-	-	
Model	All models	0.232	-	0.198	-	0.153	0.25	0.212	0.252
	Llama - Mistral	-	-	-	-	-	-	-	0.8125 (Mistral)
	Llama - Qwen	0.8451 (Qwen)	-	-	-	0.7808 (Qwen)	1.0066 (Qwen)	-	0.9184 (Qwen)
	Llama - Gemma	0.8451 (Gemma)	-	0.8451 (Gemma)	-	0.7510 (Gemma)	-	-	0.8125 (Gemma)
	Mistral - Qwen	-	-	-	-	-	-	-	-
	Mistral - Gemma	0.9518 (Gemma)	-	0.8125 (Gemma)	-	-	0.7808 (Mistral)	1.0066 (Gemma)	-
	Qwen - Gemma	-	-	0.8748 (Gemma)	-	-	-	-	-
	Llama - base	0.4683	-	-	-	0.4957	0.7683	-	-
	Mistral - base	0.9318	-	-	-	0.9199	-	0.6956	-
	Qwen - base	0.8791	0.6854	0.8791	-	0.4801	-	-	0.5987
	Gemma - base	0.517	0.543	0.9027	-	0.5495	0.7305	0.667	-

TABLE IX: Results of the statistical tests for fairness categories 5 and 6

Factor	Comparison	5. Physical Appearance				6. Race/Ethnicity			
		Acc. AMB	Acc. DIS	Bias AMB	Bias DIS	Acc. AMB	Acc. DIS	Bias AMB	Bias DIS
		Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size
All results		0.6519	-	0.2154	0.3917	0.533	0.5677	-	0.5676
Dataset	UC - UF	0.6341 (UF)	-	-	-	-	-	0.5718 (UC)	-
	UC - base	0.7656	-	-	0.6104	-	0.6713	-	0.7573
	UF - base	0.6351	-	0.2671	0.3409	0.5422	0.5469	-	0.5343
Paradigm	SFT - DPO	0.7371 (DPO)	-	0.5363 (DPO)	-	0.7302 (DPO)	0.6098 (DPO)	-	0.8561 (DPO)
	SFT - base	0.7682	-	-	0.4457	0.6156	0.6722	-	0.6719
	DPO - base	-	-	-	-	-	-	-	-
Learning Rate	1e-3 - 2e-5	0.4583 (2e-5)	-	-	-	0.5951 (2e-5)	-	-	-
	1e-3 - base	0.7443	-	-	0.4709	0.7327	0.6134	-	0.7044
	2e-5 - base	0.6519	-	-	0.3367	0.4262	0.5519	-	0.5084
Epochs	1e - 5e	-	-	0.5661 (1e)	-	-	-	-	-
	1e - base	0.7292	-	-	0.3915	0.4346	0.652	-	0.5923
	5e - base	0.6348	-	0.3083	0.3915	0.5721	0.5296	-	0.5454
PEFT Method	All methods	0.187	-	-	0.197	0.176	0.266	-	0.247
	Lora - IA ³	0.8153 (IA ³)	-	-	-	0.6582 (IA ³)	-	-	-
	IA ³ - Prompt Tuning	-	-	-	0.7159 (IA ³)	-	0.8979 (IA ³)	-	1.0113 (IA ³)
	IA ³ - P-Tuning	0.8979 (IA ³)	-	-	0.7011 (IA ³)	0.9184 (IA ³)	0.8616 (IA ³)	-	0.7620 (IA ³)
	Lora - Prompt Tuning	-	-	-	0.6582 (Lora)	-	-	-	0.6442 (Lora)
	Lora - P-Tuning	-	-	-	-	-	-	-	0.6442 (Lora)
	Prompt - P-Tuning	-	-	-	-	-	-	-	-
	Lora - base	0.7787	-	0.5457	-	0.5003	-	-	-
	IA ³ - base	0.6661	-	0.6784	-	-	-	-	-
	Prompt Tuning - base	0.6039	-	-	0.6656	0.6622	0.9523	-	1.0194
	P-Tuning - base	0.8826	-	-	0.5875	0.8123	0.8702	-	0.8123
Model	All models	0.222	0.132	-	-	0.262	-	0.335	0.265
	Llama - Mistral	-	0.8125 (Mistral)	-	-	0.7808 (Mistral)	-	0.8451 (Llama)	0.7808 (Mistral)
	Llama - Qwen	1.0066 (Qwen)	0.7879 (Qwen)	-	-	0.7808 (Qwen)	-	1.0066 (Llama)	1.0066 (Qwen)
	Llama - Gemma	-	-	-	-	1.0066 (Gemma)	-	-	-
	Mistral - Qwen	-	-	-	-	-	-	-	-
	Mistral - Gemma	-	-	-	-	-	-	-	-
	Qwen - Gemma	-	0.7808 (Qwen)	-	-	-	-	0.7808 (Gemma)	-
	Llama - base	0.5134	0.718	-	-	0.5009	0.5539	1.0333	0.8807
	Mistral - base	0.8685	0.4399	0.8533	0.4503	0.7262	0.7717	0.5653	-
	Qwen - base	-	0.4929	-	-	-	-	0.8764	0.6475
	Gemma - base	0.8792	0.802	0.5752	0.9027	0.517	0.724	0.7504	0.6358

TABLE X: Results of the statistical tests for fairness categories 7 and 8

Factor	Comparison	7. Religion				8. Socio-Economic Status			
		Acc. AMB	Acc. DIS	Bias AMB	Bias DIS	Acc. AMB	Acc. DIS	Bias AMB	Bias DIS
		Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size	Effect Size
All results		0.7453	0.4103	-	-	0.5641	0.3472	-	-
Dataset	UC - UF	-	-	-	-	-	-	-	-
	UC - base	0.8819	-	-	-	-	0.6299	-	-
	UF - base	0.7246	0.39	-	-	0.5799	0.2897	-	-
Paradigm	SFT - DPO	0.7926 (DPO)	-	0.5423 (DPO)	-	0.8269 (DPO)	-	0.5550 (DPO)	0.6795 (DPO)
	SFT - base	0.8485	0.4847	-	0.2802	0.6698	0.4982	-	-
	DPO - base	-	-	-	-	-	-	-	-
Learning Rate	1e-3 - 2e-5	0.4172 (2e-5)	-	-	-	0.6010 (2e-5)	-	-	-
	1e-3 - base	0.7768	0.4304	-	-	0.762	-	-	-
	2e-5 - base	0.7269	0.3902	-	-	0.4654	0.427	-	-
Epochs	1e - 5e	-	-	-	-	-	-	-	-
	1e - base	0.8394	0.5032	-	-	0.4439	0.6736	-	-
	5e - base	0.7188	0.3537	0.2943	-	0.6092	-	-	-
PEFT Method	All methods	0.403	0.37	-	-	0.266	0.277	0.144	-
	Lora - IA ³	0.8784 (IA ³)	-	-	-	0.7866 (IA ³)	-	-	-
	IA ³ - Prompt Tuning	0.8100 (IA ³)	1.0489 (IA ³)	-	-	0.6442 (IA ³)	0.8100 (IA ³)	-	-
	IA ³ - P-Tuning	1.0113 (IA ³)	-	-	-	0.9397 (IA ³)	-	-	-
	Lora - Prompt Tuning	-	0.7936 (Lora)	-	-	-	0.7620 (Lora)	-	-
	Lora - P-Tuning	-	-	-	-	-	-	-	-
	Prompt - P-Tuning	-	-	-	-	-	-	-	-
	Lora - base	0.7787	-	-	-	0.6875	-	-	-
	IA ³ - base	-	-	-	-	-	-	-	-
	Prompt Tuning - base	0.8675	0.8675	-	-	0.6039	0.7871	-	-
Model	P-Tuning - base	0.8826	0.5525	-	-	0.8465	0.5014	-	-
	All models	0.147	-	0.212	-	0.195	-	-	0.142
	Llama - Mistral	-	-	0.8748 (Llama)	-	-	-	-	-
	Llama - Qwen	0.8155 (Qwen)	-	-	-	0.7808 (Qwen)	-	-	-
	Llama - Gemma	-	-	-	-	1.0066 (Gemma)	-	-	-
	Mistral - Qwen	-	-	-	-	-	-	-	-
	Mistral - Gemma	-	-	-	-	-	-	-	-
	Qwen - Gemma	-	-	-	-	-	-	-	-
	Llama - base	0.4986	-	0.6512	-	0.4502	0.4671	-	0.5927
	Mistral - base	0.8571	-	0.7325	-	0.7717	-	0.522	-
	Qwen - base	0.8683	0.6848	0.644	-	-	-	-	0.477
	Gemma - base	0.8792	0.5954	0.6358	0.6853	0.716	0.8822	0.8033	-

TABLE XI: Results of the statistical tests for fairness category 9

Factor	Comparison	9. Sexual Orientation			
		Acc. AMB	Acc. DIS	Bias AMB	Bias DIS
		Effect Size	Effect Size	Effect Size	Effect Size
All results		0.705	0.3802	-	-
Dataset	UC - UF	-	-	-	-
	UC - base	0.7467	-	-	-
	UF - base	0.7004	0.3905	-	-
Paradigm	SFT - DPO	0.7582 (DPO)	-	-	-
	SFT - base	0.8008	0.4737	-	-
	DPO - base	-	-	0.6166	-
Learning Rate	1e-3 - 2e-5	-	-	-	-
	1e-3 - base	0.7236	0.4046	-	-
	2e-5 - base	0.7058	0.3645	-	-
Epochs	1e - 5e	-	-	-	-
	1e - base	0.7642	-	-	-
	5e - base	0.6923	0.3891	-	-
PEFT Method	All methods	0.197	0.174	0.234	-
	Lora - IA ³	0.8145 (IA ³)	-	-	-
	IA ³ - Prompt Tuning	-	0.8210 (IA ³)	-	-
	IA ³ - P-Tuning	0.8616 (IA ³)	-	-	-
	Lora - Prompt Tuning	-	0.6442 (Lora)	-	-
	Lora - P-Tuning	-	-	-	-
	Prompt - P-Tuning	-	-	-	-
	Lora - base	0.7516	-	0.478	-
	IA ³ - base	0.4771	-	-	0.5137
	Prompt Tuning - base	0.6377	0.8385	-	-
	P-Tuning - base	0.8056	0.4817	-	-
Model	All models	0.182	0.235	-	0.425
	Llama - Mistral	-	-	-	0.8223 (Mistral)
	Llama - Qwen	0.8487 (Qwen)	1.0066 (Qwen)	-	-
	Llama - Gemma	-	-	-	0.9518 (Llama)
	Mistral - Qwen	-	-	-	-
	Mistral - Gemma	-	-	-	1.0066 (Mistral)
	Qwen - Gemma	0.9941 (Qwen)	-	-	1.0066 (Qwen)
	Llama - base	0.5447	0.8803	-	-
	Mistral - base	0.803	-	-	0.694
	Qwen - base	-	-	-	-
	Gemma - base	0.8799	0.595	0.6358	1.0424

TABLE XII: Results of the overall statistical tests for the CodeAlpaca extension. Only significant results are shown ($p < 0.05$). Reported values are effect sizes; in pairwise comparisons, the parenthesized label marks the better mean (higher for safety and accuracy, lower for bias).

Comparison	Safety	Accuracy AMB	Accuracy DIS	Bias Score AMB	Bias Score DIS
CodeAlpaca - base	0.5240	0.6727	-	-	0.4446
CodeAlpaca - UF	-	-	0.4176 (CodeAlpaca)	-	-

TABLE XIII: Results of the statistical tests for CodeAlpaca safety categories. Only significant results are shown ($p < 0.05$); reported values are effect sizes.

Comparison	1. Illegal Activity	2. Child Abuse Content	3. Hate/ Harass/ Violence	5. Physical Harm	6. Economic Harm	7. Fraud / Deception	8. Adult Content	9. Political Campaigning	10. Privacy Violation
CodeAlpaca - base	0.5403	0.8726	0.7470	0.4677	0.5493	0.7972	0.5066	0.4784	0.6617
CodeAlpaca - UF	-	-	-	-	-	-	-	-	-

TABLE XIV: Results of the statistical tests for CodeAlpaca fairness categories. Only significant results are shown ($p < 0.05$). Reported values are effect sizes; in pairwise comparisons, the parenthesized label marks the better mean (higher for accuracy, lower for bias).

Category	Comparison	Acc. AMB	Acc. DIS	Bias AMB	Bias DIS
1. Age	CodeAlpaca - base	0.6066	-	-	-
	CodeAlpaca - UF	-	-	-	-
2. Disability Status	CodeAlpaca - base	0.6628	-	-	0.7123
	CodeAlpaca - UF	-	0.3895 (CodeAlpaca)	0.4658 (UF)	0.4497 (UF)
3. Gender Identity	CodeAlpaca - base	0.7097	-	-	-
	CodeAlpaca - UF	-	0.4698 (CodeAlpaca)	-	-
4. Nationality	CodeAlpaca - base	0.5810	-	-	-
	CodeAlpaca - UF	-	-	-	0.3654 (CodeAlpaca)
5. Physical Appearance	CodeAlpaca - base	0.5867	-	-	-
	CodeAlpaca - UF	-	-	-	-
6. Race / Ethnicity	CodeAlpaca - base	0.6793	0.5768	0.3786	0.6297
	CodeAlpaca - UF	-	-	-	-
7. Religion	CodeAlpaca - base	0.7162	0.4983	-	0.3686
	CodeAlpaca - UF	-	-	-	-
8. SES	CodeAlpaca - base	0.6528	-	-	-
	CodeAlpaca - UF	-	0.5621 (CodeAlpaca)	-	-
9. Sexual Orientation	CodeAlpaca - base	0.7859	-	-	-
	CodeAlpaca - UF	-	-	-	-

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